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Maximal induced matchings in K_4-free and K_5-free graphs

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ABSTRACT

An induced matching in a graph is a set of edges whose endpoints induce a 1-regular subgraph. It is known that every graph on n vertices has at most $10^{n/5} \approx 1.5849^n$ maximal induced matchings, and this bound is the best possible as any disjoint union of complete graphs K_5 forms an extremal graph. It is also known that the bound drops to $3^{n/3} \approx 1.4423^n$ when the graphs are restricted to the class of triangle-free (K_3-free) graphs. The extremal graphs, in this case, are known to be the disjoint unions of copies of $K_{3,3}$. Along the same line, we study the maximum number of maximal induced matchings when the graphs are restricted to K_5-free graphs and K_4-free graphs. We show that every K_5-free graph on n vertices has at most $6^{n/4} \approx 1.5651^n$ maximal induced matchings and the bound is the best possible obtained by any disjoint union of copies of K_4. When the graphs are restricted to K_4-free graphs, the upper bound drops to $8^{n/5} \approx 1.5158^n$, and it is achieved by the disjoint union of copies of the wheel graph W_5.
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1. Introduction

In extremal graph theory, we seek to determine the extremal values of graph parameters under given constraints. The study in the field started with Turán's Theorem in 1941 which ascertained the maximum number of edges in K_{s+1}-free graphs on n vertices. Since then, it has been one of the most studied research fields within graph theory. In 1949, Zykov [18] found the maximum number of r-cliques, $0 \leq r \leq s$, in K_{s+1}-free graphs on n vertices. He indeed generalized Turán's result as edges are nothing but cliques on two vertices. In 1965, Moon [11] determined the maximum number of maximal cliques in n-vertex graphs. In the same year, Moon together with Moser upper bounded the maximum number of maximal independent sets [12] in n-vertex graphs. For more examples of similar results, see [5,6,13,15–17].

Gupta, Raman, and Saurabh [7] showed in 2012 that for any fixed non-negative integer r, there exists a constant $c < 2$ such that every graph on n vertices has at most c^n maximal r-regular induced subgraphs. The value of c for $r = 0$ was already known by the aforementioned result by Moon and Moser [12]. Their result states that every graph on n vertices has at most $3^{n/3}$ maximal independent sets and it is attained by any disjoint union of copies of the triangle graph, i.e., K_3. To complement this, Gupta et al. in the same article determined the values of c for $r = 1$ and $r = 2$. In particular, for $r = 1$, they showed that every graph on n vertices has at most $10^{n/5} \approx 1.5849^n$ maximal induced matchings. They also proved that the upper bound is indeed achieved by any disjoint union of copies of K_5.

Since the extremal graphs given by Moon and Moser contain only copies of the triangle graph, it raised the question of what if the studied graphs are triangle-free. In 1993, Hujter and Tuza [8] showed that triangle-free graphs on n vertices

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have at most $2^{n/2} \approx 1.4143^n$ maximal independent sets and the bound is attained by any 1-regular graph. A decade later, Byskov [2] gave an algorithmic proof of the same result. A similar question arises as the extremal graphs by Gupta et al. for $r = 1$ contain only copies of K_5. Hence, it is interesting to discover the maximum number of maximal induced matchings when the graphs are K_5-free. It becomes more interesting when we see that Basavaraju et al. [1], in 2015, determined the number for the class of triangle-free graphs. Their result shows that every n-vertex triangle-free graph has at most $3^{n/3} \approx 1.4423^n$ maximal induced matchings and the upper bound is obtained by any disjoint union of copies of $K_{3,3}$. Hence, our initial desire was to determine where in the range between 1.4423^n and 1.5849^n the maximum number of maximal induced matchings of K_5-free graphs on n vertices lies.

In Section 3, we prove that every K_5-free graph on n vertices has at most $6^{n/4} \approx 1.5651^n$ maximal induced matchings, which is slightly smaller than the upper bound for general graphs. Furthermore, we show that this bound is tight and the graphs with exactly $6^{n/4} \approx 1.5651^n$ maximal induced matchings are any disjoint union of copies of K_4. Obtaining only copies of the K_4 graph as the extremal graphs, together with the fact that we had the upper bounds for general graphs, K_5-free graphs, and K_3-free graphs in hand, motivated us to investigate further and extend the result for the class of K_4-free graphs. In Section 4, we show that every K_4-free graph on n vertices has at most $8^{n/5} \approx 1.5158^n$ maximal induced matchings and the maximum number is attained by any disjoint union of copies of the wheel graph on 5 vertices, i.e., W_5.

Our proof strategy is the same in both cases. We assume that there exist graphs on n vertices with more maximal induced matchings than the claimed upper bound. Then, we prove a series of lemmas to discover the structural properties of these graphs. In the end, we conclude that such graphs do not exist and thus the desired upper bound holds. The same strategy was used in [1] to find the maximum number of maximal induced matchings in triangle-free graphs and in [10] to find maximum induced matchings in subcubic planar graphs.

Enumeration problems aim at outputting all solutions to their instances without duplication. Even a simple enumeration problem may produce a possibly exponential number of outputs. Therefore, the efficiency of the enumeration algorithms is often evaluated by output-polynomiality. Polynomial delay algorithms spend polynomial time in the size of the graph between the two successive outputs of an enumeration problem. It should also take polynomial time to compute the first solution and to detect that there are no further solutions. There exist polynomial delay algorithms to list the maximal independent sets of a given graph [9,14]. Together with the aforementioned combinatorial upper bound by Moon and Moser, this implies that the maximal independent sets of an n-vertex graph can be listed in time $O^*(3^{n/3}) = O(1.4423^n)$. Then, the upper bound by Hujter and Tuza guarantees that all the maximal independent sets of triangle-free graphs on n vertices can be listed faster and in time $O^*(2^{n/2}) = O(1.4143)$.

Our combinatorial upper bounds imply that algorithms listing all maximal induced matchings as their subroutine run faster in the classes of K_5-free graphs and K_4-free graphs. Cameron [3] remarked that every induced matching in a graph G is an independent set of the vertices in $[L(G)]^2$, the square graph of the line graph of G. Combining this with any algorithm that lists maximal independent sets yields an algorithm to list maximal induced matchings. Therefore, the upper bound by Gupta et al. for $r = 1$, yields polynomial delay algorithms that list all maximal induced matchings of n-vertex graphs in time $O^*(10^{n/5}) = O(1.5849^n)$. Then, our results promise faster polynomial delay algorithms to list all maximal induced matchings when the input graphs are restricted to the classes of K_5-free graphs and K_4-free graphs. We present this as corollaries of our main results, Theorems 11 and 26.

Corollary 1. *For every K_5-free graph on n vertices, all its maximal induced matchings can be listed in time $O^*(6^{n/4}) = O(1.5651^n)$ with polynomial delay.*

Corollary 2. *For every K_4-free graph on n vertices, all its maximal induced matchings can be listed in time $O^*(8^{n/5}) = O(1.5158^n)$ with polynomial delay.*

2. Definitions and notations

All graphs we consider in this article are finite, simple, and undirected. We refer the reader to Diestel [4] for terminologies and notations not defined in this section.

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. The *degree* of a vertex v which is denoted by $d(v)$ is the number of edges incident with v. We use $\delta(G)$ and $\Delta(G)$ to denote, respectively, the minimum and maximum degrees in G. For a vertex $v \in V(G)$, we use $N_G[v]$ and $N_G(v)$ to denote the closed and open neighborhoods of v, respectively. Let $A \subseteq V(G)$. The closed neighborhood of A is defined as $N_G[A] = \bigcup_{v \in A} N_G[v]$, and the open neighborhood of A is $N_G(A) = N_G[A] \setminus A$. An *induced subgraph* of G is a graph whose vertex set A is a subset of $V(G)$ and its edge set consists of all edges of G with both endpoints in A. We write $G[A]$ to denote the subgraph of G induced by A, and we write $G - A$ to denote the graph $G[V(G) \setminus A]$. If $A = \{v\}$, then we simply write $G - v$ instead of $G - \{v\}$.

For any non-negative integer r, we say that G is *r-regular* if the degree of every vertex in G is r. A 3-regular graph is called *cubic*. A *path* P_n on n vertices is a graph whose vertices can be ordered $v_1, v_2, \ldots, v_n$ such that the edges are $\{v_1v_2, \ldots, v_{n-1}v_n\}$. A *cycle* C_n on n vertices is constructed by adding edge v_nv_1 to P_n. A *complete graph* is a graph in which each pair of vertices is connected by an edge. We denote a complete graph on n vertices by K_n. The complete graph on 3 vertices is also called a *triangle*. A graph G is called *H-free* if H is not an induced subgraph of G.

A *matching* in G is a subset $M \subseteq E(G)$ such that no two edges in M share an endpoint. For a matching M in G and a vertex $v \in V(G)$, we say that M *covers* v if v is an endpoint of an edge in M. An *induced matching* M in G is a matching whose endpoints induce a 1-regular subgraph. An induced matching M in G is *maximal* if there exists no induced matching M' in G such that $M \subsetneq M'$. We write M_G to denote the set of all maximal induced matchings in G. Let X and Y be two disjoint subsets of $V(G)$. We define $M_G(X, Y)$ to be the set of all maximal induced matchings of G that cover no vertex of X and every vertex of Y. When there is no ambiguity we omit subscripts from the notations.

Let A be a set. A *partition* of A is a set of p non-empty and pairwise disjoint subsets $A_1, \ldots, A_p$ of A where $A = \cup_{i=1}^{p} A_i$. To denote that A is partitioned into subsets $A_1, \ldots, A_p$, we use $A = \biguplus_{i=1}^{p} A_i$.

Before we present our results on K_4-free graphs and K_5-free graphs, we prove a lemma that holds for every graph G and results in Corollaries 4 and 12 which are widely used throughout the article.

Lemma 3. *Let G be a graph and Y be a subset of vertices of G where $G[Y]$ is 1-regular. For every $X \subseteq V(G) \setminus Y$, it holds that $|M_G(X, Y)| \leq |M_{G-(X \cup N[Y])}|$.*

Proof. Let us denote $G - (X \cup N[Y])$ by G'. We first show that for every matching $M \in M_G(X, Y)$, it holds that $M \setminus E(G[Y]) \in M_{G'}$. Let $M \in M_G(X, Y)$ and $M' = M \setminus E(G[Y])$. Since M is a maximal induced matching that covers all vertices of Y and $G[Y]$ is 1-regular, all edges of $G[Y]$ belong to M. This and the fact that M is an induced matching in G covering none of the vertices of X guarantee that M' is an induced matching in G'. We show that M' is maximal. For contradiction, suppose there exists an edge $xy \in E(G')$ such that $M' \cup \{xy\}$ is an induced matching in G'. Neither x nor y belongs to the set $X \cup N[Y]$. This means in particular that there is no edge between the sets $\{x, y\}$ and Y. Hence, adding the edge xy to M yields an induced matching in G, contradicting the assumption that M is a maximal induced matching in G.

Note that two sets $M_1 \setminus E(G[Y])$ and $M_2 \setminus E(G[Y])$ are not equal if $M_1 \neq M_2$. Hence, it holds that $|M_G(X, Y)| \leq |M_{G'}|$ and the proof is complete. □

3. Maximal induced matchings in K_5-free graphs

This section is devoted to determining the maximum number of maximal induced matchings in K_5-free graphs. Within this section, we call a K_5-free graph on n vertices with more than $6^{n/4}$ maximal induced matchings a *counterexample*. Let us assume that there exists a counterexample. Then there exists a counterexample G with the minimum number of vertices. We prove a sequence of lemmas to discover the structural properties of G, and eventually conclude that graph G does not exist. This implies that the maximum number of maximal induced matchings of every K_5-free graph on n vertices is at most $6^{n/4}$. Then by presenting the extremal graphs having exactly $6^{n/4}$ maximal induced matchings, at the end of this section, we conclude that the upper bound is indeed best possible. We begin with a simple corollary of Lemma 3.

Corollary 4. *Let Y be a subset of vertices of G where $G[Y]$ is 1-regular and X be a subset of $V(G) \setminus Y$ where $X \cup Y \neq \emptyset$. It holds that $|M_G(X, Y)| \leq 6^{\frac{n-|X \cup N[Y]|}{4}}$.*

Proof. From the definitions of the sets X and Y, it is easy to see that $G-(X \cup N[Y])$ has fewer vertices than G. Therefore, $G - (X \cup N[Y])$ is not a counterexample and so has at most $6^{\frac{|V(G-(X \cup N[Y]))|}{4}}$ maximal induced matchings. By Lemma 3, we have

$$|M_G(X, Y)| \leq |M_{G-(X \cup N[Y])}| \leq 6^{\frac{|V(G-(X \cup N[Y]))|}{4}} \leq 6^{\frac{n-|X \cup N[Y]|}{4}},$$

which is the desired inequality. □

To prove most of the structural properties of G, we exploit proof by contradiction and contradict the fact that G is a counterexample graph. Throughout the proofs, we partition the set of all maximal induced matchings of G into some smaller sets. Each of these sets represents all maximal induced matchings of a subgraph of G with fewer vertices than n. To elaborate, we partition M_G in a way that

$$M_G = \biguplus_{i=1}^{p} M_G(X_i, Y_i),$$

where for every $1 \leq i \leq p$, we have $Y_i \subseteq V(G)$, induced subgraph $G[Y_i]$ is 1-regular, $X_i \subseteq V(G) \setminus Y_i$, and $X_i \cup Y_i \neq \emptyset$. It clearly holds that

$$|M_G| = \sum_{i=1}^{p} |M_G(X_i, Y_i)|.$$

For every $1 \leq i \leq p$, let $n_i = |X_i \cup N[Y_i]|$. By Corollary 4,

$$|M_G| = \sum_{i=1}^{p} |M_G(X_i, Y_i)| \leq \sum_{i=1}^{p} 6^{\frac{n-n_i}{4}}.$$

To get the desired contradiction, $|M_G| \leq 6^{n/4}$, we only need to show that the sum of these upper bounds is not larger than $6^{n/4}$, i.e.,

$$\sum_{i=1}^{p} 6^{\frac{n-n_i}{4}} \leq 6^{n/4}\,. \tag{1}$$

Proving this inequality needs some arguments which are similar in most of the lemmas. We explain the main idea here and refrain from reiterating similar arguments in subsequent lemmas. Observe that inequality (1) holds if polynomial $x^n - \sum_{i=1}^{p} x^{n-n_i}$ is not negative at $6^{1/4}$. Let $n_k = \max\{n_i : 1 \leq i \leq p\}$. By dividing the polynomial by x^{n-n_k}, we get a new polynomial which is called the *characteristic polynomial*. To show (1), it suffices to check that the characteristic polynomial is not negative at $6^{1/4}$. We are now ready to present our structural lemmas about G.

Lemma 5. *The graph G is connected and has at least three vertices.*

Proof. First note that a graph with at most two vertices cannot be a counterexample. Now assume for contradiction that G is not connected. Let $G_1, G_2, \ldots, G_k$ denote the connected components of G. By the choice of G, none of the connected components of G is a counterexample. Hence, $|M_{G_i}| \leq 6^{|V(G_i)|/4}$, for each $i \in \{1, \ldots, k\}$. It implies that $|M_G| = \prod_{i=1}^{k} |M_{G_i}| \leq 6^{n/4}$, contradicting the assumption that G is a counterexample. □

Lemma 6. *The graph G has no vertices of degree* 1.

Proof. By Lemma 5, graph G is connected and $n \geq 3$. Assume for contradiction that G has a vertex u of degree 1. Let v be the neighbor of u and w a neighbor of v with $w \neq u$. All the maximal induced matchings of G are partitioned into $M_G(\emptyset, \{u\})$ and $M_G(\{u\}, \emptyset)$. Note that w is not covered by any maximal induced matching covering u. By Corollary 4, it holds that

$$|M_G| \;= |M_G(\emptyset, \{u\})| \;+\; |M_G(\{u\}, \emptyset)| \;\leq 6^{(n-3)/4} + 6^{(n-1)/4}\,.$$

By proving $|M_G| \leq 6^{n/4}$, we get the desired contradiction. To do so, it suffices to show that $6^{(n-3)/4} + 6^{(n-1)/4} \leq 6^{n/4}$. Its corresponding characteristic polynomial $x^3 - x^2 - 1$ is positive at $x = 6^{1/4}$ and thus $|M_G| \leq 6^{n/4}$. This contradicts that G is a counterexample. □

Lemma 7. *The graph G has no vertices of degree* 5 *or more.*

Proof. For contradiction, suppose there is a vertex u with $d(u) \geq 5$. We partition M_G into $M_G(\{u\}, \emptyset)$ and $M_G(\emptyset, \{u\})$. We further partition $M_G(\emptyset, \{u\})$ based on the edge incident with u they include. By Corollary 4,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{d(u)} |M_G(\emptyset, \{u, v_i\})| \leq 6^{(n-1)/4} + d(u) \cdot 6^{(n-(d(u)+1))/4}\,.$$

Note that $x = 6^{1/4}$ makes $x^{d(u)+1} - x^{d(u)} - d(u)$ positive for any value of $d(u) \geq 5$. To see this, it suffices to put $6^{1/4}$ instead of x, i.e., $6^{(d(u)+1)/4} - 6^{d(u)/4} - d(u)$ and check that the expression is positive when $d(u) \geq 5$. It implies that $|M_G| \leq 6^{n/4}$ and contradicts the fact that G is a counterexample. Hence, G contains no vertices of degree 5 or more. □

Lemma 8. *The graph G has no vertices of degree* 2.

Proof. Assume for contradiction that there exists a vertex u of degree 2 in G. Let v and w be the neighbors of u. It holds that $V(G) \geq 4$, since otherwise we get a contradiction due to $|M_G| \leq 6^{n/4}$. Let us assume that both v and w have some neighbors in $V(G) \setminus N[u]$. Note that v and w may share some of these neighbors. We partition the set of all maximal induced matchings of G into the following sets: $M_G(\{u\}, \emptyset)$, $M_G(\emptyset, \{u, v\})$, and $M_G(\emptyset, \{u, w\})$. By Corollary 4,

$$|M_G| = |M_G(\{u\}, \emptyset)| + |M_G(\emptyset, \{u, v\})| + |M_G(\emptyset, \{u, w\})| \leq 6^{(n-1)/4} + 2 \cdot 6^{(n-4)/4}\,.$$

The fact that $x^4 - x^3 - 2$ is positive at $x = 6^{1/4}$ implies that $|M_G| \leq 6^{n/4}$. Therefore, G is not a counterexample, a contradiction. Hence, one of v and w has no neighbor in $V(G) \setminus N[u]$. Without loss of generality, let w be that vertex. Note that $vw \in E(G)$ since $\delta(G) \geq 2$ by Lemma 6. By Lemma 7 and the fact that v has a neighbor in $V(G) \setminus N[u]$, the degree of v is 3 or 4. Therefore, we consider two cases:

Case 1. $|N(v)| = 3$.

Let x be the neighbor of v in $V(G) \setminus N[u]$. In this case, we partition M_G based on the fact that at least one of the edges uv, uw, vw, and vx should be in any maximal induced matching of G. By Corollary 4,

$$|M_G| = |M_G(\emptyset, \{u, w\})| + |M_G(\emptyset, \{v, u\})| + |M_G(\emptyset, \{v, w\})| + |M_G(\emptyset, \{v, x\})| \leq 6^{(n-3)/4} + 2 \cdot 6^{(n-4)/4} + 6^{(n-5)/4}\,.$$

Due to Lemma 6, x has a neighbor which does not belong to $\{u, v, w\}$. This guarantees the latest inequality,

$$|M_G(\emptyset, \{v, x\})| \leq 6^{(n-5)/4}.$$

Since $x^5 - x^2 - 2x - 1$ is positive at $x = 6^{n/4}$, we have $|M_G| \leq 6^{n/4}$. This contradicts that G is a counterexample.

Case 2. $|N(v)| = 4$.

Let x and y be the two neighbors of v in $V(G) \setminus N[u]$. We partition M_G into the following five sets: $M_G(\emptyset, \{u, w\})$, $M_G(\emptyset, \{u, v\})$, $M_G(\emptyset, \{v, w\})$, $M_G(\emptyset, \{v, x\})$, and $M_G(\emptyset, \{v, y\})$. By Corollary 4,

$$|M_G| = |M_G(\emptyset, \{u, w\})| + \sum_{t \in \{u,w,x,y\}} |M_G(\emptyset, \{v, t\})| \leq 6^{(n-3)/4} + 4 \cdot 6^{(n-5)/4}.$$

Since $x^5 - x^2 - 4$ is positive at $x = 6^{1/4}$, we have $|M_G| \leq 6^{n/4}$ which completes the proof by contradicting that G is a counterexample. □

So far, we have shown that counterexample G might only include vertices of degree 3 and 4. The next lemma proves that G can only be a cubic graph.

Lemma 9. *The graph G has no vertices of degree 4.*

Proof. For contradiction, suppose G has a degree-4 vertex u. Let v_1, v_2, v_3 and v_4 denote the neighbors of u. We show that G has more than five vertices. Let G be on five vertices. Since G is not isomorphic to K_5, it has at most nine edges. It means $|M_G| \leq 6^{5/4}$ which is a contradiction to the fact that G is a counterexample. Hence, G has more than five vertices. It means that at least one of v_1, v_2, v_3, and v_4 has a neighbor in $V(G) \setminus N[u]$. To complete the proof, we check all possibilities.

First, we suppose at least two of the neighbors of u have some neighbors in $V(G) \setminus N[u]$ and get a contradiction. We partition M_G into five subsets $M_G(\{u\}, \emptyset)$ and $M_G(\emptyset, \{u, v_i\})$, where $1 \leq i \leq 4$. By Corollary 4,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{4} |M_G(\emptyset, \{u, v_i\})| \leq 6^{(n-1)/4} + 2 \cdot 6^{(n-6)/4} + 2 \cdot 6^{(n-5)/4}.$$

Note that the upper bound above is the largest among the three upper bounds which are achieved in the three cases where two, three, or four of the neighbors of u have some neighbors in $V(G) \setminus N[u]$. It is easily checked that $x^6 - x^5 - 2x - 2$ is positive when $x = 6^{1/4}$ and therefore $|M_G| \leq 6^{n/4}$. This is the desired contradiction.

So we consider the case that only one of the neighbors of u has some neighbors in $V(G) \setminus N[u]$ and get a contradiction. We assume, without loss of generality, that v_1 is that neighbor. Assume that v_1 has at least two neighbors in $V(G) \setminus N[u]$. By Corollary 4, partitioning of M_G into those covering u and those not covering it gives us:

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{4} |M_G(\emptyset, \{u, v_i\})| \leq 6^{(n-1)/4} + 6^{(n-7)/4} + 3 \cdot 6^{(n-5)/4}.$$

Since $x^7 - x^6 - 3x^2 - 1$ is positive at $x = 6^{1/4}$, we have $|M_G| \leq 6^{n/4}$. This contradicts that G is a counterexample.

Hence, it only remains to check the case that v_1 has exactly one neighbor in $V(G) \setminus N[u]$ and get a contradiction. Let us denote this neighbor of v_1 by x. Due to Lemma 8, vertex x has two neighbors other than v_1. These two neighbors do not belong to $N(u)$ as v_1 is the only vertex in $N(u)$ which has a neighbor in $V(G) \setminus N[u]$. Having $3 \leq d(v_1) \leq 4$ implies that v_1 has either one or two neighbors in $N(u)$. So, accordingly, we consider two cases:

Case 1. v_1 is adjacent to one vertex in $N(u)$.

Without loss of generality, let v_2 be that neighbor in $N(u)$. We partition M_G for those covering v_1 and those not covering it. By Corollary 4,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + \sum_{t \in \{u, v_2, x\}} |M_G(\emptyset, \{v_1, t\})| \leq 6^{(n-1)/4} + 3 \cdot 6^{(n-6)/4}.$$

Note that due to Lemma 8 both v_3 and v_4 are of degree 3, they have no neighbors in $V(G) \setminus N[u]$, and v_1 is not adjacent to them as it has only one neighbor in $N[u]$ and it is v_1. Therefore, v_3 and v_4 are adjacent to v_2 and thus not covered by induced matchings covering $\{v_1, v_2\}$. Hence, $|M_G(\emptyset, \{v_1, v_2\})| \leq 6^{(n-6)/4}$. Also, the two neighbors of x which do not belong to $N[u]$ guarantee $|M_G(\emptyset, \{v_1, x\})| \leq 6^{(n-6)/4}$. Since $x^6 - x^5 - 3$ is positive at $x = 6^{1/4}$, we have $|M_G| \leq 6^{n/4}$. This provides us with the desired contradiction.

Case 2. v_1 is adjacent to two vertices in $N(u)$.

Suppose, without loss of generality, v_1 is adjacent to v_2 and v_3. We again partition based on vertex v_1. By Corollary 4,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + \sum_{t \in \{u, v_2, v_3, x\}} |M_G(\emptyset, \{v_1, t\})| \leq 6^{(n-1)/4} + 4 \cdot 6^{(n-6)/4}.$$

Lemma 8, the assumption that none of v_2, v_3, and v_4 has a neighbor in $V(G)\setminus N[u]$, and that v_1 has only two neighbors, v_2 and v_3, within $N(u)$ imply that v_4 is adjacent to both v_2 and v_3. It is easy to see that both v_3 and v_4 are covered neither by the induced matchings covering $\{v_1, v_2\}$ nor by the induced matchings covering $\{v_1, v_3\}$ and therefore $|M_G(\emptyset, \{v_1, t\})| \leq 6^{(n-6)/4}$ when $t \in \{v_2, v_3\}$. Having $x^6 - x^5 - 4$ positive at $x = 6^{1/4}$ guarantees $|M_G| \leq 6^{n/4}$. This contradiction completes the proof. □

Note that a cubic complete graph is isomorphic to K_4. Therefore, the next lemma proves that counterexample G can only be isomorphic to K_4.

Lemma 10. *There is no non-adjacent pair of vertices in G.*

Proof. Assume for the sake of contradiction that there exists a non-adjacent pair of vertices in G. Let u be an arbitrary vertex in G with neighbors v_1, v_2, and v_3. There must exist some non-adjacent pairs of vertices in $N(u)$, since otherwise $G = K_4$ and has no non-adjacent pair of vertices. We assume without loss of generality that v_1 and v_2 form a non-adjacent pair in $N(u)$.

We first show that both v_1 and v_2 are adjacent to v_3. To do so, again without loss of generality, we assume that v_1 is not adjacent to v_3. Since $d(v_1) = 3$, it has two neighbors in $V(G) \setminus N[u]$. We partition all maximal induced matchings of G into the following four sets: $M_G(\{u\}, \emptyset)$ and $M_G(\emptyset, \{u, v_i\})$, where $1 \leq i \leq 3$. By Corollary 4,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{3} |M_G(\emptyset, \{u, v_i\})| \leq 6^{(n-1)/4} + 6^{(n-6)/4} + 2 \cdot 6^{(n-5)/4}.$$

The reason why $|M_G(\emptyset, \{u, v_i\})| \leq 6^{(n-5)/4}$, for $i \in \{2, 3\}$, is that each of v_2 and v_3 has at least one neighbor in $V(G) \setminus N[u]$ as they are of degree 3 and not adjacent to v_1. Since $x^6 - x^5 - 2x - 1$ is positive at $x = 6^{1/4}$, we conclude that $|M_G| \leq 6^{n/4}$. This contradicts the existence of G and concludes that both v_1 and v_2 are adjacent to v_3.

Having $\{v_1v_3, v_2v_3\} \in E(G)$ implies that each of v_1 and v_2 has a third neighbor in $V(G) \setminus N[u]$. Note that they might share their third neighbors. Let us denote by x the neighbor of v_1 in $V(G) \setminus N[u]$. Note also that v_3 is adjacent to both v_1 and v_2. We partition M_G into $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{v_1, u\})$, $M_G(\emptyset, \{v_1, v_3\})$, and $M_G(\emptyset, \{v_1, x\})$. By Corollary 4,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + \sum_{t \in \{u, v_2, x\}} |M_G(\emptyset, \{v_1, t\})| \leq 6^{(n-1)/4} + 2 \cdot 6^{(n-5)/4} + 6^{(n-6)/4}.$$

Note that x has two neighbors which do not belong to $\{u, v_1, v_3\}$. This guarantees that $|M_G(\emptyset, \{v_1, x\})| \leq 6^{(n-6)/4}$. One can easily check that $x^6 - x^5 - 2x - 1$ is positive at $x = 6^{1/4}$ and therefore $|M_G| \leq 6^{n/4}$. This is a contradiction to the definition of G. Hence, there is no non-adjacent pair of vertices in G. □

Finally, we are ready to present the main theorem of this section.

Theorem 11. *Every K_5-free graph on n vertices contains at most $6^{n/4}$ maximal induced matchings, and this bound is attained by any disjoint union of copies of K_4.*

Proof. So far, we have shown that G can only be isomorphic to K_4. Clearly, K_4 has exactly 6 maximal induced matchings which is equal to $6^{n/4}$ as $n = |V(G)| = 4$. This contradicts the assumption that G is a counterexample. Hence, there exists no counterexample graphs and every K_5-free graph on n vertices has at most $6^{n/4}$ maximal induced matchings.

It remains to show that the upper bound is best possible. Let G be the disjoint union of p copies of K_4 for some positive integer p. Every maximal induced matching in G contains exactly one edge of each connected component of G, which implies that $|M_G| = 6^p = 6^{n/4}$. This completes the proof.

4. Maximal induced matchings in K_4-free graphs

This section is devoted to proving that every K_4-free graph on n vertices has at most $8^{n/5}$ maximal induced matchings and this bound is best possible. From now on, we call a K_4-free graph on n vertices with more than $8^{n/5}$ maximal induced matchings a *counterexample*. Let us assume that there exists a counterexample and, similar to the previous section, we assume that among all counterexamples G is one with the minimum number of vertices. By the definition of counterexample graphs, $|M_G| > 8^{n/5}$. We again discover a sequence of structural properties of G to conclude that G does not exist. During the proofs of the lemmas, we will partition the set all of maximal induced matchings of G into some sets in a way that the sum of their cardinalities is upper bounded by $8^{n/5}$, the desired inequality. To do so, we will check the corresponding characteristic polynomials to be positive at $8^{1/5}$. Similar to Section 3, we begin with a simple corollary of Lemma 3.

Corollary 12. *Let Y be a subset of vertices of G where $G[Y]$ is 1-regular and X be a subset of $V(G) \setminus Y$ where $X \cup Y \neq \emptyset$. It holds that $|M_G(X, Y)| \leq 8^{\frac{n-|X \cup N[Y]|}{5}}$.*

Proof. From the definitions of sets X and Y, it is easy to see that $G-(X\cup N[Y])$ has fewer vertices than G. Therefore, $G-(X\cup N[Y])$ is not a counterexample and so has at most $8^{\frac{|V(G-(X\cup N[Y]))|}{5}}$ maximal induced matchings. By using Lemma 3, we get

$$|M_G(X,Y)| \leq |M_{G-(X\cup N[Y])}| \leq 8^{\frac{|V(G-(X\cup N[Y]))|}{5}} \leq 8^{\frac{n-|X\cup N[Y]|}{5}},$$

which is the desired inequality. □

Lemma 13. *The graph G is a connected graph on at least three vertices.*

Proof. First note that a graph with at most two vertices cannot be a counterexample as it contains at most one edge and thus at most one maximal induced matching. Now assume for contradiction that G is not connected and $G_1, G_2, \ldots, G_k$ are its connected components. By the choice of G, none of the connected components of G is a counterexample. Hence, $|M_{G_i}| \leq 8^{|V(G_i)|/5}$, for each $i \in \{1,\ldots,k\}$. It implies that $|M_G| = \prod_{i=1}^{k} |M_{G_i}| \leq 8^{n/5}$, which is a contradiction to the fact that G is a counterexample. □

Lemma 14. *The graph G has no vertices of degree 1.*

Proof. By Lemma 13, G is connected and has at least three vertices. Assume for contradiction that G has a vertex u of degree 1. Let v be the only neighbor of u. All maximal induced matchings of G are partitioned into $M_G(\emptyset,\{u\})$ and $M_G(\{u\},\emptyset)$. Note that $N(v)\setminus\{u\}$ is not empty and none of its vertices is covered by maximal induced matchings covering u. By Corollary 12, we have

$$|M_G| = |M_G(\emptyset,\{u\})| + |M_G(\{u\},\emptyset)| \leq 8^{(n-3)/5} + 8^{(n-1)/5}.$$

By proving $|M_G| \leq 8^{n/5}$, we get the desired contradiction. To do so, it suffices to show that $8^{(n-3)/5} + 8^{(n-1)/5} \leq 8^{n/5}$. The corresponding characteristic polynomial is $x^3 - x^2 - 1$ and it is positive at $x = 8^{1/5}$. It implies that $|M_G| \leq 8^{n/5}$. This is a contradiction as G is assumed to be a counterexample. Hence, G has no vertices of degree 1. □

Lemma 15. *The graph G has no vertices of degree 6 or more.*

Proof. Suppose for the sake of contradiction that u is a vertex in G with $d(u) \geq 6$. We partition M_G into $M_G(\{u\},\emptyset)$ and $M_G(\emptyset,\{u\})$. Based on which incident edge with u is included, $M_G(\emptyset,\{u\})$ is further partitioned as follows:

$$M_G(\emptyset,\{u\}) = \biguplus_{v\in N(u)} M_G(\emptyset,\{u,v\}).$$

By Corollary 12,

$$|M_G| = |M_G(\{u\},\emptyset)| + \sum_{v\in N(u)} |M_G(\emptyset,\{u,v\})| \leq 8^{(n-1)/5} + d(u)8^{(n-(d(u)+1))/5}.$$

Its characteristic polynomial $x^{d(u)+1} - x^{d(u)} - d(u)$ is positive at $8^{1/5}$ for any value of $d(u) \geq 6$. To see this, it suffices to put $8^{1/5}$ instead of x, i.e., $8^{(d(u)+1)/5} - 8^{d(u)/5} - d(u)$ and check that the expression is positive when $d(u) \geq 6$. It implies that $|M_G| \leq 8^{n/5}$ and therefore G is not a counterexample. This contradiction guarantees that G has no vertices of degree 6 or more. □

We now aim to show that G cannot have a vertex of degree 2. We first need the following supporting lemma.

Lemma 16. *Let u be a vertex of degree 2 in G. Each neighbor of u has exactly one neighbor in $V(G)\setminus N[u]$.*

Proof. Let us assume that G has a degree-2 vertex u with two neighbors v and w. We first show that both v and w have some neighbors in $V(G)\setminus N[u]$. To do so, we assume for contradiction that at least one of v and w has no such neighbors. Let, without loss of generality, v be the vertex with no neighbors in $V(G)\setminus N[u]$. Lemma 14 implies that $vw \in E(G)$. Vertex w has some neighbors in $V(G)\setminus N[u]$, since otherwise, G is a triangle and not a counterexample. We partition maximal induced matchings of G into $M_G(\{w\},\emptyset)$ and $M_G(\emptyset,\{w\})$. The latter is further partitioned as follows:

$$M_G(\emptyset,\{w\}) = \biguplus_{x\in N(w)} M_G(\emptyset,\{x,w\}).$$

Note that those maximal induced matchings not covering w contain uv since otherwise they are not maximal. Therefore, $M_G(\{w\},\emptyset) = M_G(\{w\},\{u,v\})$. By Corollary 12,

$$|M_G| = |M_G(\{w\},\{u,v\})| + \sum_{x\in N(w)} |M_G(\emptyset,\{x,w\})| \leq 8^{(n-3)/5} + d(w)8^{(n-(d(w)+1))/5}.$$

The characteristic polynomial, $x^{d(w)+1} - x^{d(w)-2} - d(w)$, is positive at $x = 8^{1/5}$ for any value of $d(w)$. Therefore, the inequality $|M_G| \leq 8^{n/5}$ holds. This contradicts the assumption that G is a counterexample, and thus, both v and w have some neighbors in $V(G) \setminus N[u]$.

Now we show that each of v and w has exactly one neighbor apart from those in $N[u]$. For this purpose, we consider two separate cases.

Case 1. $vw \in E(G)$.

We assume without loss of generality that $d(w) \geq d(v)$. We partition the maximal induced matchings of G into partition sets $M_G(\{u\}, \emptyset)$, $M_G(\emptyset, \{u, v\})$, and $M_G(\emptyset, \{u, w\})$. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + |M_G(\emptyset, \{u, v\})| + |M_G(\emptyset, \{u, w\})| \leq 8^{(n-1)/5} + 8^{(n-(d(v)+1))/5} + 8^{(n-(d(w)+1))/5}.$$

We identify the values of $d(v)$ and $d(w)$ for which

$$8^{(n-1)/5} + 8^{(n-(d(v)+1))/5} + 8^{(n-(d(w)+1))/5} < 8^{n/5}.$$

By Lemmas 14 and 15, every vertex of G has a degree between 2 and 5. It is easy to see that its characteristic polynomial $x^{d(w)+1} - x^{d(w)} - x^{d(w)-d(v)} - 1$ is positive at $x = 8^{1/5}$ for all values of $d(v)$ and $d(w)$ except when $d(w) = d(v) = 3$. It implies that $d(w) = d(v) = 3$ as G is a counterexample. Hence, each of v and w is of degree 3 and thus has exactly one neighbor in $V(G) \setminus N[u]$.

Case 2. $vw \notin E(G)$.

We assume again that $d(w) \geq d(v)$. We partition the maximal induced matchings of G exactly the way we did in Case 1. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + |M_G(\emptyset, \{u, v\})| + |M_G(\emptyset, \{u, w\})| \leq 8^{(n-1)/5} + 8^{(n-(d(v)+2))/5} + 8^{(n-(d(w)+2))/5}.$$

Similar to Case 1, we identify the values of $d(v)$ and $d(w)$ for which the desired inequality holds. The corresponding characteristic polynomial, $x^{d(w)+2} - x^{d(w)+1} - x^{d(w)-d(v)} - 1$, is positive at $x = 8^{1/5}$ for all values of $d(v)$ and $d(w)$ except when $d(w) = d(v) = 2$. It implies that $d(w) = d(v) = 2$ as G is a counterexample. Hence, each of v and w is a degree-2 vertex with exactly one neighbor in $V(G) \setminus N[u]$. □

Lemma 17. *The graph G has no vertices of degree 2.*

Proof. Assume for contradiction that G has a degree-2 vertex u. Let v and w be the neighbors of u. Due to Lemma 16 each of v and w has exactly one neighbor in $V(G) \setminus N[u]$. Let us call these neighbors of v and w, respectively, p and q. Note that p and q might denote the same vertex if v and w share their neighbors in $V(G) \setminus N[u]$. We differentiate between two cases; $vw \in E(G)$ and $vw \notin E(G)$.

Case 1. $vw \in E(G)$.

In this case $d(v) = d(w) = 3$. We first show that both p and q are of degree at least three. Let us assume that this is not the case. If p and q denote the same vertex whose degree is two, G consists of four vertices $\{u, v, w, p\}$ and five edges $\{uv, uw, vw, vp, wp\}$, i.e., it is a diamond. This graph has five maximal induced matchings. Since $5 < 8^{4/5}$, this contradicts the fact that G is a counterexample. Hence, when p and q are the same vertex, its degree is at least three. Therefore, it remains to show that when p and q are two separate vertices, each of them is of degree at least three. We assume without loss of generality that $d(p) = 2$. Since p is not adjacent to u and w, it is a degree-2 vertex having a neighbor, namely v, with two neighbors in $V(G) \setminus N[p]$. This contradicts Lemma 16. Hence, both p and q are of degree at least 3. We are now ready to contradict our assumption that G has the degree-2 vertex u.

Let us first assume that p and q are two separate vertices. We partition the set of maximal induced matchings of G into $M_G(\emptyset, \{v, w\})$, $M_G(\{w\}, \{v\})$, $M_G(\{v\}, \{w\})$, and $M_G(\{v, w\}, \emptyset)$. Note that all matchings in $M_G(\emptyset, \{v, w\})$ contain edge vw, matchings in $M_G(\{w\}, \{v\})$ are partitioned into $M_G(\{w\}, \{v, u\})$ and $M_G(\{w\}, \{v, p\})$, matchings in $M_G(\{w\}, \{v\})$ are partitioned into $M_G(\{v\}, \{w, u\})$ and $M_G(\{v\}, \{w, q\})$, and each matching in $M_G(\{v, w\}, \emptyset)$ does not cover u. By Corollary 12,

$$\begin{aligned}|M_G| &= |M_G(\emptyset, \{v, w\})| + |M_G(\{w\}, \{v\})| + |M_G(\{v\}, \{w\})| + |M_G(\{v, w\}, \emptyset)| \\ &\leq 8^{(n-5)/5} + (8^{(n-4)/5} + 8^{(n-d(p)-3)/5}) + (8^{(n-4)/5} + 8^{(n-d(q)-3)/5}) + 8^{(n-3)/5}.\end{aligned}$$

Without loss of generality, we assume that $d(p) \geq d(q)$. It is easy to see that $x^{d(p)+3} - x^{d(p)} - 2x^{d(p)-1} - x^{d(p)-d(q)} - x^{d(p)-2} - 1$ is positive at $x = 8^{1/5}$ for any value of $d(p) \geq 2$ when $d(q)$ is either 3,4 or 5. This guarantees that $|M_G| \leq 8^{n/5}$ and G is not a counterexample.

To finish Case 1, it only remains to get a similar contradiction when p and q denote the same vertex. In such case, we partition the set of maximal induced matchings of G into $M_G(\{p\}, \emptyset)$ and $M_G(\emptyset, \{p\})$. Observe that when a maximal induced matching of G does not cover p, it includes exactly one of the edges uv, uw, and vw. Therefore, $M_G(\{p\}, \emptyset)$ is further partitioned into $M_G(\{p\}, \{u, v\})$, $M_G(\{p\}, \{u, w\})$, and $M_G(\{p\}, \{v, w\})$. We also partition $M_G(\emptyset, \{p\})$ again based on which edge incident with p they include. All these together imply by Corollary 12 that:

$$|M_G| = |M_G(\{p\}, \emptyset)| + |M_G(\emptyset, \{p\})| \leq 3 \cdot 8^{(n-4)/5} + d(p) \cdot 8^{(n-d(p)-2)/5}.$$

It is easy to see that $x^{d(p)+2} - 3 \cdot x^{d(p)-2} - d(p)$ is positive at $x = 8^{1/5}$ for all values of $3 \leq d(p) \leq 5$. It implies that $|M_G| \leq 8^{n/5}$ and contradicts the existence of G.

Case 2. $vw \notin E(G)$.

In this case $d(v) = d(w) = 2$. We first prove that G is a cycle. To do so, we assume for contradiction that G is not a cycle. Then, the path P that starts from u, goes through v, and contains only degree 2 vertices of G, contains at least two and at most $n - 2$ vertices. Ordering vertices of P starting from u, let x be the last vertex of P. Let y be the neighbor of x which does not belong to P. Clearly, y is of a degree larger than 2. Vertex x is a degree-2 vertex that has a neighbor y with at least two neighbors not in $N[x]$. The existence of x is a contradiction to Lemma 16. This proves that G is a cycle.

It is easy to see that C_3, C_4 and C_5 are not counterexamples. Consider a cycle C_n where $n \geq 6$. We partition M_G into $M_G(\emptyset, \{u\})$ and $M_G(\{u\}, \emptyset)$. Then we further partition $M_G(\emptyset, \{u\})$ into two subsets $M_G(\emptyset, \{u, v\})$ and $M_G(\emptyset, \{u, w\})$. Partition set $M_G(\{u\}, \emptyset)$ is again partitioned into four subsets $M_G(\{u, v, w\}, \emptyset)$, $M_G(\{u, w\}, \{v\})$, $M_G(\{u, v\}, \{w\})$, and $M_G(\{u\}, \{v, w\})$. Clearly, every matching in $M_G(\{u, w\}, \{v\})$ includes edge vp, every matching in $M_G(\{u, v\}, \{w\})$ includes edge wq, and every matching in $M_G(\{u\}, \{v, w\})$ includes both edges vp and wq. Note that the latter holds as $vw \notin E(G)$, the graph G is a cycle on at least 6 vertices, and four vertices v, p, w, and q induce a 1-regular subgraph of G. By Corollary 12, we have

$$\begin{aligned}|M_G| &= |M_G(\emptyset, \{u, v\})| + |M_G(\emptyset, \{u, w\})| + |M_G(\{u, v, w\}, \emptyset)| + |M_G(\{u, v\}, \{w, q\})| + |M_G(\{u, w\}, \{v, p\})| + |M_G(\{u\}, \{v, w\})| \\ &\leq 2 \cdot 8^{(n-4)/5} + 8^{(n-3)/5} + 2 \cdot 8^{(n-5)/5} + 8^{(n-6)/5}.\end{aligned}$$

Since $x^6 - x^3 - 2x^2 - 2x - 1$ is positive at $x = 8^{1/5}$, it implies that $|M_G| \leq 8^{n/5}$. This completes the proof by contradicting the existence of G. □

Before we present Lemma 20 and prove that G does not contain any vertices of degree 5, we prove two more lemmas about the structure of the neighbors of any degree-5 vertex in G.

Lemma 18. *If there exists a degree-5 vertex u in G where at least three of its neighbors have some neighbors in $V(G) \setminus N[u]$, then G is not a counterexample.*

Proof. Let u be a vertex of degree 5 in G and v_1, v_2, v_3, v_4, and v_5 be its five neighbors. We assume, without loss of generality, that each of v_1, v_2, and v_3 has at least one neighbor in $V(G) \setminus N[u]$. We partition the set of maximal induced matchings of G into $M_G(\{u\}, \emptyset)$ and $M_G(\emptyset, \{u, v_i\})$, where $1 \leq i \leq 5$. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{5} |M_G(\emptyset, \{u, v_i\})| \leq 8^{(n-1)/5} + (3 \cdot 8^{(n-7)/5} + 2 \cdot 8^{(n-6)/5}).$$

Since $x^7 - x^6 - 2x - 3$ is positive at $x = 8^{1/5}$, we deduce that $|M_G| \leq 8^{n/5}$. Hence, G is not a counterexample and the proof is complete. □

Lemma 19. *There are no degree-3 vertices in the neighborhood of a degree-5 vertex in G.*

Proof. Let u be a vertex of degree 5 and v_1, v_2, v_3, v_4, and v_5 be its neighbors. Let also $d(v_1) = 3$ and $N(v_1) = \{u, x, y\}$. Note that x or y may belong to $\{v_2, v_3, v_4, v_5\}$. We consider three cases based on this.

Case 1. Both x and y belong to $V(G) \setminus N[u]$.

We partition the set of all maximal induced matchings of G into the following four subsets: $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{x, v_1\})$, and $M_G(\emptyset, \{y, v_1\})$. By Corollary 12,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{x, v_1\})| + |M_G(\emptyset, \{y, v_1\})| \leq 8^{(n-1)/5} + 8^{(n-8)/5} + 2 \cdot 8^{(n-5)/5}.$$

To see the inequality note that each of x and y has at least one neighbor not in $\{u, v_1, x, y\}$ as $\delta(G) \geq 3$. It holds that $|M_G| \leq 8^{n/5}$ as $x^8 - x^7 - 2x^2 - 1$ is positive at $x = 8^{1/5}$. Hence, G is not a counterexample, a contradiction.

Case 2. Both x and y belong to $N(u)$.

Both x and y belong to set $\{v_2, v_3, v_4, v_5\}$. Since G is K_4-free, then $xy \notin E(G)$. Then each of x and y has a neighbor which does not belong to $\{u, v_1, x, y\}$ as $\delta(G) \geq 3$. We now partition the set of all maximum induced matchings of G into $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{x, v_1\})$, and $M_G(\emptyset, \{y, v_1\})$. By Corollary 12,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{x, v_1\})| + |M_G(\emptyset, \{y, v_1\})| \leq 8^{(n-1)/5} + 8^{(n-6)/5} + 2 \cdot 8^{(n-5)/5}.$$

Since $x^6 - x^5 - 2x - 1$ is positive at $x = 8^{1/5}$, we have $|M_G| \leq 8^{n/5}$ which is a contradiction to the existence of G.

Case 3. One of x and y belongs to $N(u)$ while the other one does not.

Without loss of generality, we assume that x belongs to $N(u)$ and it is the same as v_2. We first prove that each of x and y has a neighbor which is not in $\{u, v_1, x, y\}$. This is clear for the case of y as $\delta(G) \geq 3$. We show that x has such a neighbor. Note that since G is K_4-free, at least two of the vertices v_3, v_4, and v_5 are not adjacent. Say v_3 and v_4 are those two vertices. Assume that neither v_3 nor v_4 is adjacent to x. Then both v_3 and v_4 have a neighbor in $V(G) \setminus N[u]$ since the

minimum degree is at least 3. Hence, v_1, v_3, and v_4 are three vertices in G each of which has a neighbor in $V(G) \setminus N[u]$. This contradicts Lemma 18 and thus x is adjacent to at least one of v_3 and v_4. This means that x has a neighbor that is not in $\{u, v_1, x, y\}$.

We now partition M_G into four subsets $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{x, v_1\})$, and $M_G(\emptyset, \{y, v_1\})$. Note that each of x and y has at least one neighbor which is not in $\{u, v_1, x, y\}$. By Corollary 12, we have

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{x, v_1\})| + |M_G(\emptyset, \{y, v_1\})| \leq 8^{(n-1)/5} + 8^{(n-7)/5} + 2 \cdot 8^{(n-5)/5}.$$

To see that inequality $|M_G| \leq 8^{n/5}$ holds it is enough to check that $x^7 - x^6 - 2x^2 - 1$ is positive at $x = 8^{1/5}$. This contradiction completes the proof. □

Lemma 20. *There are no vertices of degree* 5 *in* G.

Proof. Suppose for the sake of contradiction that G has a vertex u of degree 5. Let v_1, v_2, v_3, v_4, and v_5 be the five neighbors of u. Note that according to Lemma 19 each of these five neighbors is of degree at least 4. Also, due to Lemma 18, at least three of these vertices have all their neighbors within $N[u]$. Let v_1 be one of those vertices. Let us assume, again without loss of generality, that v_1 is adjacent to v_2, v_3, and v_4. There is no edge between any pair of v_2, v_3, and v_4 since G is a K_4-free graph. It implies that each of these three vertices has at most three neighbors within $N[u]$. Hence, each of them has at least one neighbor in $V(G) \setminus N[u]$ which is a contradiction to Lemma 18 and the proof is complete. □

So far, we have proven that degrees of vertices of G can only be 3 and 4. By the remaining of lemmas in this section, we show that G can only be isomorphic to the wheel graph on five vertices, W_5.

Lemma 21. *If each of the neighbors of a degree-*4 *vertex* u *has at least one neighbor in* $V(G) \setminus N[u]$*, then* G *is not a counterexample.*

Proof. Let u be a vertex of degree 4 in G where v_1, v_2, v_3, and v_4 are its neighbors. We partition the set of all maximal induced matchings of G into five subsets $M_G(\{u\}, \emptyset)$ and $M_G(\emptyset, \{u, v_i\})$, where $1 \leq i \leq 4$. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{4} |M_G(\emptyset, \{u, v_i\})| \leq 8^{(n-1)/5} + 4 \cdot 8^{(n-6)/5}.$$

Inequality $|M_G| \leq 8^{n/5}$ holds as $x^6 - x^5 - 4$ is positive at $x = 8^{1/5}$. It implies that G is not a counterexample. □

Lemma 22. *Let* v *be a degree-*3 *vertex in the neighborhood of a degree-*4 *vertex* u *in* G*. All neighbors of* v *are within* $N[u]$.

Proof. Let u be a vertex of degree 4 in G with neighbors v_1, v_2, v_3, and v_4 where v_1 is of degree 3. Let also x and y be the two other neighbors of v_1. We show that G is not a counterexample when one or both of x and y belong to $V(G) \setminus N[u]$. First, we assume that both x and y belong to $V(G) \setminus N[u]$ and show that G is not a counterexample. By Corollary 12, partitioning M_G into $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{x, v_1\})$, and $M_G(\emptyset, \{y, v_1\})$ gives us

$$|M_G| = |M_G(\{u\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{x, v_1\})| + |M_G(\emptyset, \{y, v_1\})| \leq 8^{(n-1)/5} + 8^{(n-7)/5} + 2 \cdot 8^{(n-5)/5}.$$

To see the inequality, note that $\delta(G) \geq 3$ and therefore each of x and y has at least one neighbor which does not belong to $\{u, v_1, x, y\}$. It holds that $x^7 - x^6 - 2x^2 - 1$ is positive at $x = 8^{1/5}$. Therefore, $|M_G| \leq 8^{n/5}$ contradicts the existence of G.

We now prove that G is not a counterexample if one of x and y belongs to $V(G) \setminus N[u]$. Without loss of generality, we assume that x is in $N[u]$ and it is the same as v_2. Since $\delta(G) \geq 3$, vertex y has a neighbor which is not in $\{u, v_1, x, y\}$. We show that x is adjacent to at least one of v_3 and v_4. Suppose it is not correct. Since $\delta(G) \geq 3$, each of x, v_3, and v_4 has a neighbor in $V(G) \setminus N[u]$. Hence, each of the four neighbors of u has a neighbor in $V(G) \setminus N[u]$. Due to Lemma 21, graph G has no such a degree-4 vertex. So, x is adjacent to at least one of v_3 and v_4. Hence, both x and y have at least one neighbor which is not in $\{u, v_1, x, y\}$. We partition M_G the same way we did in the previous case and get the following inequality by Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{x, v_1\})| + |M_G(\emptyset, \{y, v_1\})| \leq 8^{(n-1)/5} + 8^{(n-6)/5} + 2 \cdot 8^{(n-5)/5}.$$

Since $x^6 - x^5 - 2x - 1$ is positive at $x = 8^{1/5}$, inequality $|M_G| \leq 8^{n/5}$ holds. This implies that G is not a counterexample.

Hence, all neighbors of v_1 belong to $N[u]$ and the proof is complete. □

Lemma 23. *The neighbors of a degree-*4 *vertex are not all of degree* 4.

Proof. Let us assume for contradiction that u is of degree 4 and all its neighbors v_1, v_2, v_3, and v_4 are of degree 4. Due to Lemma 21, at least one of v_1, v_2, v_3, and v_4 has all its neighbors in $N[u]$. Let v_1 be such a vertex which is adjacent to all v_2, v_3, and v_4. Since G is a K_4-free graph, no three vertices in $N(u)$ form a triangle and therefore there is no edge in

$G[\{v_2, v_3, v_4\}]$. This means that each of these three vertices has exactly two neighbors in $V(G) \setminus N[u]$. We partition M_G into those covering u and those not covering it. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{4} |M_G(\emptyset, \{u, v_i\})| \leq 8^{(n-1)/5} + 8^{(n-5)/5} + 3 \cdot 8^{(n-7)/5}.$$

Inequality $|M_G| \leq 8^{n/5}$ holds since $x^7 - x^6 - x^2 - 3$ is positive at $x = 8^{1/5}$. This completes the proof by contradicting that G is a counterexample. □

Lemma 24. *If there exists a degree-4 vertex in G then none of its neighbors is of degree 4.*

Proof. Let u be a vertex of degree 4 with neighbors v_1, v_2, v_3, and v_4. Let us assume for contradiction that u has a neighbor of degree 4 and let v_1 be that vertex. Due to Lemma 23, there is at least one vertex of degree 3 in $N(u)$. Let us assume without loss of generality that v_2 is that vertex. To complete the proof, we check the two possibilities $v_1v_2 \in E(G)$ and $v_1v_2 \notin E(G)$ and get a contradiction in each case.

Case 1. $v_1v_2 \in E(G)$.

Due to Lemma 22, all neighbors of v_2 are in $N(u)$. Let v_3 be the third neighbor of v_2. Since G is K_4-free then $v_1v_3 \notin E(G)$. Observe that v_1 is a degree-4 vertex in G which has a degree-3 neighbor, namely v_2, with a neighbor, v_3, in $V(G) \setminus N[v_1]$. This contradicts Lemma 22.

Case 2. $v_1v_2 \notin E(G)$.

Due to Lemma 22, all neighbors of v_2 are in $N[u]$. The other two neighbors of v_2 are v_3 and v_4. Note that $v_3v_4 \notin E(G)$ since otherwise, there would be a K_4 in G. Vertex v_3 is not of degree 4 since otherwise, its degree-3 neighbor v_2 would have a neighbor, v_4, which belongs to $V(G) \setminus N[v_3]$, a contradiction to Lemma 22. Similarly, v_4 is not of degree 4. Hence, $d(v_3) = d(v_4) = 3$. Again due to Lemma 22, we have $v_1v_3, v_1v_4 \in E(G)$. Since $v_1v_2 \notin E(G)$, vertex v_3 (similarly v_4) is a degree-3 vertex in the neighborhood of a degree-4 vertex, namely v_1, which one of its neighbors, v_2, belongs to $V(G) \setminus N[v_1]$. This contradicts Lemma 22. □

The next lemma together with Lemmas 22 and 24 implies that G can only be isomorphic to W_5 graph.

Lemma 25. *The graph G has a vertex of degree 4.*

Proof. Let us assume for contradiction that G is cubic. Let u be any vertex in G and v_1, v_2, and v_3 be its neighbors. The neighborhood of u has three possible structures, as G is K_4-free. We study each of these structures as different cases.

Case 1. $G[N(u)]$ has no edges.

In this case, we partition M_G into $M_G(\{u\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{u, v_2\})$, and $M_G(\emptyset, \{u, v_3\})$. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{3} |M_G(\emptyset, \{u, v_i\})| \leq 8^{(n-1)/5} + 3 \cdot 8^{(n-6)/5}.$$

Since plugging $x = 8^{1/5}$ makes $x^6 - x^5 - 3$ positive, $|M_G| \leq 8^{n/5}$ and thus G is not a counterexample.

Case 2. $G[N(u)]$ has one edge.

Without loss of generality, let $v_1v_2 \in E(G)$. We partition the set of all maximal induced matchings of G into $M_G(\{u\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{u, v_2\})$, and $M_G(\emptyset, \{u, v_3\})$. By Corollary 12,

$$|M_G| = |M_G(\{u\}, \emptyset)| + \sum_{i=1}^{3} |M_G(\emptyset, \{u, v_i\})| \leq 8^{(n-1)/5} + 2 \cdot 8^{(n-5)/5} + 8^{(n-6)/5}.$$

Since $x^6 - x^5 - 2x - 1$ is positive at $x = 8^{1/5}$, we have $|M_G| \leq 8^{n/5}$ which contradicts the fact that G is a counterexample.

Case 3. $G[N(u)]$ has two edges.

Let us assume without loss of generality that $v_1v_2, v_2v_3 \in E(G)$. Then $v_1v_3 \notin E(G)$. Let x be the third neighbor of v_1, which is not in $N[u]$. In this case, M_G is partitioned a bit differently into $M_G(\{v_1\}, \emptyset)$, $M_G(\emptyset, \{u, v_1\})$, $M_G(\emptyset, \{v_1, v_2\})$, and $M_G(\emptyset, \{x, v_1\})$. By Corollary 12,

$$|M_G| = |M_G(\{v_1\}, \emptyset)| + |M_G(\emptyset, \{u, v_1\})| + |M_G(\emptyset, \{v_1, v_2\})| + |M_G(\emptyset, \{x, v_1\})| \leq 8^{(n-1)/5} + 2 \cdot 8^{(n-5)/5} + 8^{(n-6)/5}.$$

To see the inequality note that x has at least one neighbor which does not belong to $\{u, v_1, v_2, v_3\}$. Since $x^6 - x^5 - 2x - 1$ is again positive at $x = 8^{1/5}$, inequality $|M_G| \leq 8^{n/5}$ holds. This contradiction completes the proof and there should exist a degree-4 vertex in G. □

By Lemma 25, if counterexample G exists it includes a degree-4 vertex u. Then, Lemma 24 guarantees that all four neighbors of u are of degree 3 and their neighbors belong to $N[u]$ due to Lemma 22. Combining these together with the fact that G is K_4-free implies that our counterexample cannot be any graph other than the wheel graph on five vertices, W_5. We are now ready to present the main theorem that upper bounds the number of maximal induced matchings in K_4-free graphs. The proof is similar to the proof of Theorem 11.

Theorem 26. *Every K_4-free graph on n vertices has at most $8^{n/5}$ maximal induced matchings, and this bound is obtained by any disjoint union of copies of W_5.*

Proof. We have shown so far that G can only be isomorphic to W_5. Clearly, W_5 has exactly eight maximal induced matchings which is equal to $8^{n/5}$ as $n = |V(G)| = 5$. This contradicts the assumption that G is a counterexample. Hence, there exists no counterexample and thus every K_4-free graph on n vertices has at most $8^{n/5}$ maximal induced matchings.

It remains to show that the upper bound is best possible. Let H be the disjoint union of p copies of W_5 for some positive integer p. Every maximal induced matching in H contains exactly one edge of each connected component of H, which implies that $|M_G| = 8^p = 8^{n/5}$. This completes the proof. □

Data availability

No data was used for the research described in the article.

References

[1] M. Basavaraju, P. Heggernes, P. van 't Hof, R. Saei, Y. Villanger, Maximal induced matchings in triangle-free graphs, J. Graph Theory 83 (3) (2015) 231–250, http://dx.doi.org/10.1002/jgt.21994.
[2] J.M. Byskov, Enumerating maximal independent sets with applications to graph colouring, Oper. Res. Lett. 32 (2004) 547–556, http://dx.doi.org/10.1016/j.orl.2004.03.002.
[3] K. Cameron, Induced matchings, Discrete Appl. Math. 24 (1989) 97–102, http://dx.doi.org/10.1016/0166-218X(92)90275-F.
[4] R. Diestel, Graph Theory, Electronic Edition, Springer-Verlag, Heidelberg, 2005, http://dx.doi.org/10.1007/978-3-662-53622-3.
[5] J. Eckhoff, The maximum number of triangles in a K4-free graph, Discrete Math. 194 (1–3) (1999) 95–106, http://dx.doi.org/10.1016/S0012-365X(98)00120-4.
[6] D.C. Fisher, Lower bounds on the number of triangles in a graph, J. Graph Theory 13 (4) (1989) 505–512, http://dx.doi.org/10.1002/jgt.3190130411.
[7] S. Gupta, V. Raman, S. Saurabh, Maximum r-regular induced subgraph problem: Fast exponential algorithms and combinatorial bounds, SIAM J. Discrete Math. 26 (2012) 1758–1780, http://dx.doi.org/10.1137/09077850X.
[8] M. Hujter, Zs. Tuza, The number of maximal independent sets in triangle-free graphs, SIAM J. Discrete Math. 6 (1993) 284–288, http://dx.doi.org/10.1137/0406022.
[9] D.S. Johnson, M. Yannakakis, C.H. Papadimitriou, On generating all maximal independent sets, Inform. Process. Lett. 27 (3) (1988) 119–123, http://dx.doi.org/10.1016/0020-0190(88)90065-8.
[10] R.J. Kang, M. Mnich, T. Müller, Induced matchings in subcubic planar graphs, SIAM Journal on Discrete Mathematics 26 (3) (2012) 1383–1411, http://dx.doi.org/10.1137/100808824.
[11] J.W. Moon, On the number of complete subgraphs of a graph, Can. Math. Bull. 8 (1965) 831–834, http://dx.doi.org/10.4153/CMB-1965-065-9.
[12] J.W. Moon, L. Moser, On cliques in graphs, Israel J. Math. 3 (1965) 23–28, http://dx.doi.org/10.1007/BF02760024.
[13] J.H. Spencer, On cliques in graphs, Israel J. Math. 9 (1971) 419–421, http://dx.doi.org/10.1007/BF02771457.
[14] S. Tsukiyama, M. Ide, H. Ariyoshi, I. Shirakawa, A new algorithm for generating all the maximal independent sets, SIAM J. Comput. 6 (3) (1977) 505–517, http://dx.doi.org/10.1137/0206036.
[15] H. Wang, On the maximum number of independent cycles in a graph, Discrete Math. 205 (1–3) (1999) 183–190, http://dx.doi.org/10.1016/S0012-365X(99)00009-6.
[16] H.S. Wilf, The number of maximal independent sets in a tree, SIAM J. Algebr. Discrete Methods 7 (1) (1986) 125–130, http://dx.doi.org/10.1137/0607015.
[17] G.C. Ying, K.K. Meng, B.E. Sagan, V.R. Vatter, Maximal independent sets in graphs with at most r cycles, J. Graph Theory 53 (4) (2006) 270–282, http://dx.doi.org/10.1002/jgt.20185.
[18] A.A. Zykov, On some properties of linear complexes, Mat. Sb. (N.S.) 24 (66) (1949) 163–188 (Russian).